Name : Nitrogen

Type : Trojan

Description

Nitrogen Trojan which is a type of malware which was first reported in 2023, the Nitrogen primarily intended for financial gain and data theft. Usually, phishing or malvertising efforts are used to spread it, tricking victims into downloading a malicious ZIP file from a hacked website. The Trojan masquerades as a Python binary and installs itself on the system after it has been run. Nitrogen will create a backdoor once it is operational, giving attackers constant access to the compromised system. After that, it releases other malicious tools, such Cobalt Strike and Sliver, which provide attackers more access privileges and network navigation capabilities. The trojan's main advantage is its ability to be unnoticed, as it hides its actions from antivirus software by exploiting Py-Fuscate. Nitrogen also targets sensitive data, such as bank information and login passwords, and may finally result in the deployment of BlackCat ransomware. The files on the compromised machine are encrypted by this last payload, which then requests a ransom to unlock them. Nitrogen poses a serious risk because of its capacity to travel across networks and go undetected, especially in businesses without sophisticated security procedures. **[1]**

Protection methods Against Nitrogen Trojan:

1. Prevent downloading files from untrusted sources particularly free software**[2]**
2. Avoid opening suspicious emails containing attachments **[2]**
3. Ensure Multi factor authentication for an extra layer of protection. **[2]**
4. Regularly update all software and operating systems to fix vulnerabilities. **[2]**
5. Limit the spread of malware by segmenting the network. **[2]**

How to Cope if Infected:

1. Disconnect the machine immediately from the network to prevent further spread.**[2]**
2. Perform a comprehensive malware scan using security tools to detect and remove the Trojan. **[2]**
3. Update passwords for all important accounts connected to important systems to prevent further access for the attackers**[2]**
4. In large-scale infections, seek assistance from a cybersecurity expert.**[2]**

Name : Akira

Type : Ransomware

Description

The Akira Ransomeware was first reported in March 2023 and has been well known for its multi extortion tactics, which targets significant organisations in industries like manufacturing, healthcare, education, and banking. Akira puts additional pressure on victims who must deal with both encryption and possible data exposure by threatening to publish stolen data on a TOR based website if the ransom is not paid.**[3]** Usually, the attack starts by using spear phishing techniques, flaws in multi-factor authentication (MFA), or vulnerabilities in publicly accessible services like VPN software. To ensure successful encryption, once inside, the attackers increase their privileges and utilise programs like PowerShell to turn off volume shadow copies. The .akira extension is applied to encrypted files.**[4]** Akira also engages in data exfiltration to further pressure victims. Stolen data is uploaded to the group's TOR website, adding another layer of extortion where victims must pay to prevent the publication of their sensitive files. This tactic significantly increases the financial damage of the attack. **[4]**

How to Protect Against Akira Ransomware

1. Educate staff members about phishing and dubious attachments. **[4]**
2. To stop unwanted access, enable MFA for every account. **[4]**
3. Update and patch systems frequently to fix vulnerabilities to prevent attackers from exploiting them.**[4]**
4. Establish robust password policies and change passwords frequently. **[4]**
5. Make regular backups of your data and make sure they are safely kept. **[4]**

How to Cope if Infected

1. Disconnect from the network to stop and control further damage. **[4]**
2. Run a full malware scan from a trusted software and remove the ransomware. **[4]**
3. Restore files from clean backups if available . **[4]**

Name : Raptor Train

Type : Botnet

Description:

Flax Typhoon hacking group, linked to the People's Republic of China, was behind the advanced botnet malware called Raptor Train which was first reported in July 2023.The botnet mainly targets consumer IoT devices that owners frequently leave unprotected, like routers, IP cameras, DVRs, and NAS equipment. These gadgets join a massive botnet that the attackers may remotely control. Raptor Train is used for denial-of-service (DDoS) assaults, data theft, and cyber-espionage. Integrity Technology Group created a unique tool called KRLab, which is used by the malware to function. Attackers can use this technology to remotely take over compromised computers and carry out a number of nefarious tasks, such as data theft and surveillance.Because of its secret design, which uses regular internet traffic as cover, the botnet is hard for conventional security measures to identify. Once compromised, devices can be used for espionage and cyberattacks on valuable organizations, such as government agencies and vital infrastructure.**[5]**

How to protect against Raptor Train Botnet

1. Close unused ports to prevent vulnerabilities from being exploited by the attacker **[6]**
2. Segment network to reduce the spread of botnet infections as a layer of extra protection**[6]**
3. Utilise strong firewalls to block and detect botnets**[6]**
4. Regularly update programmes to patch vulnerabilities and ensure protection .**[6]**

How to cope if infected

1. Get a cybersecurity professional to help address and amend the issues and get expertise.**[7]**
2. Factory reset the infected device to get rid of the botnets**[7]**
3. Monitor network to look for unusual activities **[7]**

Name : HotPage

Type : backdoor adware

Description

HotPage was first reported in 2024, mainly targeting Windows systems in China. At first it shows itself as an “Internet café security solution,” claiming to block ads and improve the browsing experience. However, its real purpose is to inject unwanted ads, track user activity, and gather sensitive data. HotPage operates by using a kernel mode driver to gain system-level privileges and avoid being detected by most security softwares. When executed,it embeds itself in the system, creating registry entries or adding itself to startup folders so it can automatically restart after a reboot. HotPage increases its privileges by exploiting vulnerabilities within the system, giving it deeper access to critical areas of the operating system. Then a command and control channel is being set up, enabling attackers to give commands remotely and get data from the compromised system. The malware conducts keylogging, takes screenshots, and records videos to capture sensitive information, including login credentials, financial data, and personal details. This stolen data is transmitted back to the attackers through encrypted communication channels, to avoid detection from traditional security methods. **[8]**

How to protect against HotPage

1. Utilise trusted sources when downloading applications to avoid adware bundled with freeware **[10]**
2. Use antivirus with real-time scanning to detect and block suspicious processes during installation for extra protection. **[10]**
3. Use extensions to block pop-ups and malicious ads **[10]**
4. Restrict user permissions to prevent adware from installing deeply in the system. **[10]**

5. Educate users to read, recognize and avoid suspicious updates and installations **[10]**

How to cope if infected

1. Check systems installed programs and uninstall unwanted or suspicious programs**[9]**
2. Restore backed up data from a clean backup**[9]**
3. Run an antivirus scan and follow the instructions given to solve the problem on PC's**[9]**
4. Attempt a factory reset on affected tablet or phones**[9]**

Name : Tiktok Malware Scam
Type : Trojan

Description

The TikTok Malware Scam was first reported in 2024 and targeted global TikTok users, particularly in Vietnam. It exploits user trust by impersonating real TikTok accounts or influencers. The scam starts with fraudulent messages and the victims are tricked into downloading a malicious version of the TikTok app, disguised as an update,claiming the user has been selected for an exclusive beta test of a new version of TikTok. The scammers promise new features to entice users. Sometimes attackers impersonate popular influencers to increase the fact that they are real. Once installed, the Trojan is activated and works by silently running in the background of the device, granting attackers unauthorized access to the victim's data. It steals login credentials, financial information, and personally identifiable information. The malware also enables cybercriminals to monitor sensitive activities, giving them control over the victim's accounts.This combination of social engineering and malware distribution makes the scam particularly dangerous for TikTok users.**[11]**

How to protect against TikTok Malware Scam

1. Prevent downloading or installing softwares from untrusted sources**[12]**
2. Avoid opening suspicious attachments in an email from unknown senders**[12]**
3. Ensure all software is updated with the latest updates.**[12]**
4. Ensure that trojan antivirus is being installed and run on the laptop**[12]**

How to cope with it

1. Contact an IT professional to help with the trojan issues **[13]**
2. Disconnect the machine from the internet to reduce damage and scan system using antivirus**[13]**
3. Update the database of the that helps the antivirus software to identify new threats**[13]**

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